

ShopEasy E-Commerce Platform

Client / BA Requirements Package for QA Testing

Version: 1.0 **Purpose:** Training / realistic QA project simulation

This package is written as if the client and BA are providing requirements to the QA team. It is intended to be sufficient for preparing a Test Plan, Test Scenarios, Test Cases and RTM.

Reference note: E-commerce requirements commonly separate functional requirements (what the system does) from non-functional requirements (quality attributes such as performance, security and usability), and include integrations, business rules and acceptance criteria.

1. Project Overview

Project: ShopEasy E-Commerce Platform

Application: Web-based e-commerce application

Primary users: Guest Customer, Registered Customer, Admin

Target platforms: Desktop and mobile browsers

2. Business Objective

The client wants an online platform where customers can register, browse and search products, manage a cart, complete checkout and payment, place and track orders, request cancellation/returns/refunds, receive notifications and review purchased products. Administrators must manage products, inventory, customers, coupons and orders.

3. Functional Requirements Specification (FRS)

FR-001 — User Registration

- New customers can create an account using first name, last name, email, mobile number, password and confirm password.
- Email must be unique; mobile number must be valid.
- Password: minimum 8 characters, at least one uppercase, one number and one special character.
- Password and confirmation must match.
- Successful registration sends an account confirmation notification.

FR-002 — Customer Login

- Registered customers can log in using registered email and password.
- Valid credentials allow access; invalid credentials show an appropriate error.
- Password is masked.
- Account is temporarily locked after 5 consecutive failed login attempts.
- Forgot Password is available.
- Successful login redirects to the home/account page.

FR-003 — Product Browsing

- Customers can view categories, subcategories, product lists, images, names, prices, availability and product details.

FR-004 — Product Search

- Search supports product name, keyword and SKU. No-match searches display an appropriate 'No products found' message.

FR-005 — Product Filtering & Sorting

- Filters: category, brand, price range, rating and availability. Sorting: price low-high, high-low, rating and newest.

FR-006 — Product Details

- Display name, image, description, price, discount, final price, stock, SKU, rating and reviews. Out-of-stock products cannot be added to cart.

FR-007 — Shopping Cart

- Customer can add, remove, increase and decrease quantities and continue shopping.
- Quantity cannot exceed available stock.
- Cart shows product, quantity, unit price, discount and subtotal.
- Cart total recalculates when quantity/items change.
- Customer is informed if a cart product becomes unavailable.

FR-008 — Wishlist

- Logged-in customers can add/remove products and move wishlist items to cart.

FR-009 — Address Management

- Customer can add, edit, delete and select a default delivery address. Address includes name, mobile, address lines, city, state, PIN, country.

FR-010 — Checkout

- Checkout covers cart review, delivery address, shipping method, coupon, payment and order confirmation. Display product, quantity, price, discount, shipping, tax and final payable amount.

FR-011 — Coupon

- Valid coupons apply discounts. Invalid/expired coupons are rejected. Minimum-order rules are enforced. Discount must be reflected in final amount.

FR-012 — Payment

- Support credit card, debit card, UPI, net banking and wallet. Successful payment creates/continues order confirmation; failed payment must not falsely confirm the order.

FR-013 — Order Placement

- After successful payment, create a unique Order ID, reduce inventory, store order details, display confirmation and send notification.

FR-014 — Order History

- Customer can view previous orders with ID, date, products, amount, payment status and order status.

FR-015 — Order Tracking

- Status progresses: Confirmed → Processing → Shipped → Out for Delivery → Delivered. Customer can view current status.

FR-016 — Order Cancellation

- Customer can cancel an order before shipment. Once Shipped, cancellation is not allowed.

FR-017 — Return

- Eligible delivered products can be returned within 7 days of delivery. Reasons: damaged, wrong product, defective or other.

FR-018 — Refund

- After approved return and product receipt, refund is initiated and completed according to applicable payment/order rules.

FR-019 — Notifications

- Notifications are sent for registration, password reset, order confirmation, payment success/failure, shipping, delivery, cancellation, return and refund.

FR-020 — Product Review

- Only customers who purchased a product can review it. Rating is 1–5; customer can add/edit/view review.

FR-021 — Admin Product Management

- Admin can add, edit, deactivate/delete products, upload images, update price/description/stock and assign categories.

FR-022 — Admin Inventory Management

- Admin can view inventory, change stock, identify low stock and mark products unavailable.

FR-023 — Admin Order Management

- Admin can view/search orders, view details, update status and process cancellation, return and refund activities.

FR-024 — Admin User Management

- Admin can view/search customers, activate/deactivate accounts and view customer order history.

FR-025 — Admin Coupon Management

- Admin can create/edit/activate/deactivate coupons and configure expiry, minimum order amount and discount percentage.

4. Non-Functional Requirements (NFR)

NFR-001 — Performance

Normal pages should load within 3 seconds under agreed test conditions; search results within 2 seconds; checkout API within 3 seconds.

NFR-002 — Availability

Target availability is 99.9%, excluding approved maintenance windows.

NFR-003 — Security

Use HTTPS; mask passwords; protect authentication; prevent unauthorized access; validate inputs; use secure session management; protect against common web attacks.

NFR-004 — Browser Compatibility

Support Chrome, Edge, Firefox and Safari.

NFR-005 — Responsive Design

Application should work on desktop, tablet and mobile.

NFR-006 — Usability

Customer should be able to Search → Add to Cart → Checkout → Pay without unnecessary or confusing steps.

NFR-007 — Scalability

For this training project, target 5,000 concurrent users.

5. Business Rules

ID	Rule
BR-001	Email must be unique during registration.
BR-002	Account locks after 5 consecutive failed login attempts.
BR-003	Product quantity cannot exceed available stock.
BR-004	Out-of-stock products cannot be purchased.
BR-005	Coupon minimum-order rules must be enforced where configured.
BR-006	Expired coupons cannot be used.
BR-007	Order is created only after successful payment.
BR-008	Inventory is reduced after successful order placement.
BR-009	Order cannot be cancelled after shipment.
BR-010	Return is allowed within 7 days of delivery.
BR-011	Only purchased products can be reviewed.
BR-012	Admin-only functions must not be accessible to customers.

6. User Roles

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Guest	Browse/search products
Customer	Purchase products, manage orders, reviews
Admin	Manage products, inventory, orders, users
Payment Gateway	Process payment
Notification Service	Send email/SMS notifications

7. Integrations

- Payment Gateway
- Email Service
- SMS/Notification Service
- Database
- Shipping/Delivery service (if applicable)

8. Data Requirements

Major entities expected: Customer, Address, Wishlist, Cart, Order, Order Item, Payment, Shipment, Return/Refund, Product, Category, Inventory, Review and Coupon.

9. Acceptance Criteria — Key Examples

Login: Given a registered customer, when valid credentials are entered, then the customer successfully logs in.

Add to Cart: Given an in-stock product, when Add to Cart is selected, then the product appears in the cart with correct quantity and price.

Checkout: Given an available product in cart, when checkout is completed, then the correct order amount is displayed.

Payment: Given valid checkout information, when payment succeeds, then an order is created and confirmation is displayed.

Cancellation: Given an order that has not shipped, when the customer requests cancellation, then the order is cancelled and applicable refund processing begins.

10. Testing Dependencies / Documents Required from Client or BA

Document	Why QA needs it
BRD	Business objectives, business scope and high-level requirements.
FRS/SRS	Detailed functional requirements and expected system behavior.
NFR	Performance, security, availability, usability and scalability requirements.
User Stories & Acceptance Criteria	Agreed user behavior and definition of done/acceptance.
UI/UX Designs	Expected screens, fields, navigation and visual behavior.
API Specification	Endpoints, request/response, status codes and validation rules.
Database / ER Diagram	Data relationships and persistence expectations.
Test Environment Details	QA/staging URL, versions, access and infrastructure dependencies.

Document	Why QA needs it
Test Data	Customer, product, order, payment and other required data.
Payment Gateway Sandbox	Test credentials, supported methods and callback behavior.
Notification Specification	Email/SMS templates, triggers and expected content.
Browser/Device Matrix	Supported browser and device combinations.
Release/Deployment Plan	Build versions, deployment dates and rollback/dependency information.
Security Requirements	Applicable security controls, roles and authorization expectations.
Performance Requirements	Load targets, response-time thresholds and test conditions.

11. QA Exercise — What You Should Produce

Using this client requirement package, prepare the following in order:

Step	Deliverable	What you do
1	Test Plan	Define scope, inclusions, exclusions, strategy, environment, defects, roles, schedule, deliverables, entry/exit criteria, suspension/resumption criteria, tools, risks and approvals.
2	Test Scenarios	Derive high-level scenarios from FRs and business rules. Do not jump directly to hundreds of test cases.
3	Test Cases	Convert scenarios into detailed positive, negative, boundary and validation test cases.
4	RTM	Map BRD/FRS requirements to scenarios and test cases so every requirement has coverage.
5	Test Data	Identify data needed for each important flow.
6	Execution	Execute against the agreed environment and record pass/fail/block results.
7	Defects	Raise defects with reproducible steps, expected/actual result, severity, priority and evidence.
8	Closure	Prepare test summary/closure outputs and obtain required sign-off.

12. Recommended Requirement Flow

Client Business Need → BRD / User Story → FRS/SRS → NFR / Business Rules → Acceptance Criteria → Test Plan → Test Scenarios → Test Cases → RTM → Execution → Defects → Retest/Regression → Test Closure

Important: This is a realistic training specification, not a production client's confidential document. Thresholds such as 3-second response time and 5,000 concurrent users are intentionally defined for this exercise and should be replaced by approved project values in a real engagement.

Sources / reference context

The structure of this training package follows common e-commerce requirements practice: functional and non-functional requirements, integrations, business rules, user roles, interfaces and acceptance criteria. Current e-commerce requirements guidance also emphasizes defining integrations and checkout/order rules early.

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